

13.1. Standard problem. Solve the initial value problem

$$y''(x) + 4y'(x) + 4y(x) = 2e^{-2x}, \quad y(0) = y'(0) = 0.$$

“Solve” means two things: 1) write down a solution/a family of solutions and check that indeed they work, 2) explain why such solution/family of solutions is unique (you may refer to the notes).

13.2. Constant coefficient ODEs. Solve the following differential equations, find first the most general solution and then impose the initial conditions.

1. $2y'' - 24y' + 72y = 0, \quad y(0) = 1, \quad y'(0) = 4,$

2. $y'' + y' - 6y = 0, \quad y(0) = 1, \quad y'(0) = 7,$

3. $y'' - 2y' + 2y = 0, \quad y(0) = 0, \quad y'(0) = -1.$

What changes if we allow complex-valued solutions?

13.3. Non-homogeneous problems.

1. Determine the general solution of the differential equation

$$y'' - 4y' + 4y = \sin(x).$$

2. Determine the solution of the initial value problem

$$y'' - y = x, \quad y(0) = 1, \quad y'(0) = 3.$$

3. Find the maximally defined solution of the initial value problem

$$y' - \left(\frac{4}{x} + 1\right)y = x^4, \quad y(1) = 1.$$

Hint: use the integrating factor trick. That is, look at the equation solved by $u(t) := e^{H(t)}y(t)$, with H satisfying $H'(t) = \frac{4}{t} + 1$.

13.4. Cauchy-Lipschitz. Put these differential equations in normal form, i.e. find in each case $U \subset \mathbb{R}^2$ and $F: U \rightarrow \mathbb{R}$ such that

$$y'(x) = F(x, y(x)), \quad (x, y) \in U \subset \mathbb{R}^2.$$

Then explain in each case whether Cauchy-Lipschitz-Picard theorem applies.

Finally find an explicit formula for the solution $y(x) = \dots$ for a general initial datum.

1. $(x^2 - x)y' = y^2 + y,$

2. $y' + e^y = 1,$

3. $xy' = 1 - y^2,$

4. $y'x^2 = y^2 + yx + x^2$.

13.5. Multiple choice. Check the correct answer(s).

1. For which of the following differential equations does the set of all solutions form a vector space?
 - (a) $y' - 2y = 0$.
 - (b) $x(y')^2 - 2y = 0$.
 - (c) $y' - x\sqrt{|y|} = 0$.
 - (d) $y' - y = 1$.
 - (e) $y' - x^2y = 0$.
2. Let $A \in \text{Mat}_{n,n}(\mathbb{R})$ be a real $n \times n$ matrix. Which of the following statements about the solutions of the linear differential equation system $\dot{x} = Ax$ are generally true?
 - (a) Every solution $x: \mathbb{R} \rightarrow \mathbb{R}^n$ with $\lim_{t \rightarrow \infty} \|x(t)\|_2 = \infty$ grows exponentially as $t \rightarrow \infty$.
 - (b) There exists a nontrivial constant solution $x: \mathbb{R} \rightarrow \mathbb{R}^n$ if and only if A has a nontrivial kernel.
 - (c) If A is real diagonalizable and there exists a nontrivial bounded solution $x: \mathbb{R} \rightarrow \mathbb{R}^n$, then 0 is an eigenvalue of A .
 - (d) If A is complex diagonalizable and there exists a nontrivial bounded solution $x: \mathbb{R} \rightarrow \mathbb{R}^n$, then 0 is an eigenvalue of A .
3. Let $U \subset \mathbb{R} \times \mathbb{R}^n$ be open, $f: U \rightarrow \mathbb{R}^n$ locally Lipschitz continuous in space and $(t_0, x_0) \in U$. Then the initial value problem

$$\dot{x}(t) = f(t, x(t)), \quad x(t_0) = x_0,$$

has, by the Picard–Lindelöf theorem, there is an open interval $I_{\max} \subset \mathbb{R}$ and a maximal solution $x: I_{\max} \rightarrow \mathbb{R}^n$ whose graph is contained in U . Which of the following statements generally hold for the Picard iteration for this initial value problem?

- (a) There exists a $\delta > 0$ such that the Picard iteration on $I = [t_0 - \delta, t_0 + \delta]$ converges uniformly to $x|_I$.
- (b) The Picard iteration converges uniformly to $x|_{I_{\max}}$ on $I_{\max}$.
- (c) The Picard iteration converges uniformly to $x|_K$ on every compact subinterval K of $I_{\max}$.

13.6. True or False. Say whether the following statements are true or false:

1. Let $A(t) \in C^1(\mathbb{R}, \mathbb{R}^{n \times n})$ be a time-dependent **symmetric** matrix and $x_0 \in \mathbb{R}^n$. Then $x(t) := \exp A(t)x_0$ solves the ODE

$$x'(t) := A'(t)x(t), \quad x(0) = 0?$$

2. Let $A(t), B(t) \in C^1(\mathbb{R}, \mathbb{R}^{n \times n})$ be time-dependent matrices. Then

$$(AB)'(t) = A'(t)B(t) + A(t)B'(t).$$

3. Let $A(t), B(t) \in C^1(\mathbb{R}, \mathbb{R}^{n \times n})$ be time-dependent matrices. Then

$$\exp(A(t) + B(t)) = \exp(A(t)) \cdot \exp(B(t)).$$

4. Let $A, B \in \mathbb{R}^{n \times n}$ be matrices, which are symmetric and positive definite. Then

$$\exp(A + B) = \exp(A) \cdot \exp(B).$$